

Data Dictionary — Sentinel-1 SAR Backscatter

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

Source: Sentinel-1 GRD (COPERNICUS/S1_GRD) via Google Earth Engine
Native resolution: 10 m (IW mode) · **Revisit:** ~6 days (combined S1A + S1B)
Period: 2019-10-01 → 2024-07-31 (5 wheat seasons) · **Parcels:** 1,500
Configuration: IW mode, VV + VH polarisations, DESCENDING orbit only
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1. Overview

This dataset provides Sentinel-1 Synthetic Aperture Radar (SAR) backscatter values for 1,500 winter wheat parcels in Centre-Val de Loire, France. Unlike Sentinel-2 (optical), SAR observes the Earth through clouds and at night, providing a continuous all-weather complement to the optical signal.

Five yearly CSV files are delivered, one per wheat campaign, in **raw scene-by-scene format** (no pre-computed composites). This allows downstream computation of any custom temporal aggregation (3-day, 5-day, 7-day, 15-day, etc.) directly in Python.

2. File inventory

Filename	Approx size	Rows (incl. NaN)	Period
sentinel1_sar_raw_2020.csv	25 MB	601,500	2019-10 → 2020-07
sentinel1_sar_raw_2021.csv	~25 MB	~600,000	2020-10 → 2021-07
sentinel1_sar_raw_2022.csv	~25 MB	~600,000	2021-10 → 2022-07
sentinel1_sar_raw_2023.csv	~25 MB	~600,000	2022-10 → 2023-07
sentinel1_sar_raw_2024.csv	~25 MB	~600,000	2023-10 → 2024-07

3. Column dictionary

KEY	identifiant	META	metadata	FEATURE	backscatter value
Column	Type	Unit	Description		
parcelle_uid	KEY	str	—		Unique observation ID (year_IDPARCEL). Primary join key with weather, soil, target.
year	KEY	int	year		Harvest year (2020–2024).
scene_date	META	date	YYYY-MM-DD		Acquisition date of the Sentinel-1 scene.
VV	FEATURE	float	dB		Backscatter coefficient σ^0 in VV polarisation (vertical transmit, vertical receive). Typical range for vegetated wheat: [-15, -5] dB.
VH	FEATURE	float	dB		Backscatter coefficient σ^0 in VH polarisation (vertical transmit, horizontal receive — cross-pol). Typical range: [-22, -12] dB.
VV_VH_ratio	FEATURE	float	dB		VV/VH ratio = VV – VH (in dB). Sensitive to canopy structure. Typical range: [4, 12] dB.

NaN structure (similar to Sentinel-2 raw): ~80 % of rows have NaN values. This is **not missing data** — GEE produces one row per (parcel × scene_date), but each scene only covers part of the study area. Parcels outside the scene footprint receive NaN. Use `df.dropna(subset=['VV'])` to filter to real measurements (~120,000 valid per year, median 100 scenes per parcel per season).

4. SAR backscatter — physical interpretation

4.1 What σ^0 measures

The backscatter coefficient σ^0 measures the fraction of microwave energy reflected back to the satellite sensor by the Earth's surface. It is expressed in decibels (dB) on a logarithmic scale, with typical values for land surfaces in [-30, 0] dB. Higher (less negative) values indicate stronger backscatter, usually from rough surfaces, dense vegetation or wet surfaces.

4.2 VV vs VH polarisations

Polarisation	Mechanism	Sensitivity
VV (co-pol)	Surface scattering, double-bounce	Surface roughness, soil moisture, vertical canopy structures (e.g. cereal stems)
VH (cross-pol)	Volume scattering	Canopy biomass, dense vegetation volume
VV/VH ratio	Depolarisation indicator	Canopy density (decreases as canopy thickens)

4.3 Wheat phenology in SAR

Period	VV (dB)	Phase	Physical mechanism
Oct–Nov	~-9	Sowing/tillering	Rough bare soil + small plants
Dec–Feb	~-9	Winter dormancy	Partial cover, low biomass
Mar	~-11	Stem elongation	Canopy develops, soil signal attenuates
Apr–May	~-13,5	Heading/flowering	Dense canopy → maximum attenuation
Jun	~-12	Maturation	Senescence begins, attenuation decreases
Jul	variable	Harvest	Canopy removed → strong VV recovery

5. Acquisition geometry

Parameter	Value	Rationale
Instrument mode	IW (Interferometric Wide)	Standard mode for terrestrial observation, 250 km swath, 5×20 m resolution
Polarisations kept	VV and VH	The two transmitted polarisations available in IW mode over land
Orbit pass	DESCENDING only	Consistent viewing geometry — ASCENDING has different incidence angles
Resolution filter	10 m (resolution_meters)	High-resolution product (GRD-HR)
Output unit	dB (decibels)	Native GEE output for the GRD collection — no further log transformation needed

6. Pre-processing applied

- **Calibration:** applied automatically by GEE during ingestion. σ^0 values are radiometrically calibrated and in dB.
- **Speckle filter:** a `focal_median 50 m` filter is applied per scene to reduce granular SAR noise (speckle). This is a moderate, robust smoothing that preserves edges.
- **VV/VH ratio:** computed as `vv - vh` (subtraction in dB = ratio in linear).
- **Parcel-level aggregation:** for each scene, the mean of valid pixels under the parcel polygon is taken (`reduceRegions, scale=30 m`).

7. Data quality notes

- **Temporal density:** ~100 scenes per parcel per season — over twice the density of Sentinel-2 (~42). Excellent for sequence models.
- **Uniform coverage:** SAR is unaffected by clouds, so density is constant year-round (~8 scenes/parcel/month) — unlike Sentinel-2 which drops to 2-3/month in winter.
- **Speckle residual:** after median filtering, individual scenes still show some pixel-to-pixel variability (typical ± 1 -2 dB). Temporal aggregation (composite N days) reduces this further.

- **Orbit choice:** using only DESCENDING means ~half the available scenes were discarded for consistency. Adding ASCENDING would double density but introduce bias from different incidence angles.

8. Project usage

- **Join key:** `parcelle_uid` matches the parcel sample, weather, soil and target datasets.
- **Computing composites in Python:**

```
import pandas as pd
raw = pd.read_csv('sentinel1_sar_raw_2020.csv', parse_dates=['scene_date'])
raw = raw.dropna(subset=['VV'])
raw['window'] = raw['scene_date'].dt.floor('7D')
comp_7d = raw.groupby(['parcelle_uid', 'window']).agg({
    'VV': ['median', 'mean', 'max', 'min', 'std'],
    'VH': ['median', 'mean', 'max', 'min', 'std'],
    'VV_VH_ratio': ['median', 'mean', 'max', 'min', 'std']
}).reset_index()
```

- **Suggested derived features:** `vv_min_doy`, `vv_min_value`, `vv_amplitude`, `vv_post_min_slope`, `ratio_vvvh_recovery_doy` (harvest marker), `vh_integral_AMJ`. See EDA report for details.
- **Complementarity with S2:** S1 fills the winter gap left by clouds in Sentinel-2 — combined density reaches ~18 scenes/parcel/month year-round.